

Lab 2, Act 3 – Answer Key

How much should you trust the ‘best’ fit?

NRES 746 – Instructor Answer Key (not for student distribution)

Part 1: Plan the bootstrap

1a. Similar resolution (50-60 values per parameter) for 3 parameters:

```
60^3
```

```
## [1] 216000
```

Roughly 200,000 combinations – three orders of magnitude more than Act 1’s 2D grid, and each one requires evaluating the SSR at 15 points. Still technically possible but slow and wasteful; this is the motivation for `optim()`. (Some students may estimate using 50 instead of 60 – either order of magnitude, ~125,000-200,000, is an acceptable answer.)

1b. A correct plain-English bootstrap algorithm should include:

1. Set the number of bootstrap replicates (e.g. 200+).
2. For each replicate: draw a sample of 15 observations, *with replacement*, from the full set of 15 raw tank observations (pooled, not stratified by tank size).
3. Refit both models (`fit_ricker()` and `fit_flex_ricker()`) to that resampled data, using the resampled x and y vectors as input.
4. Store the fitted parameters from both models for that replicate.
5. After all replicates are done, look at the distribution (e.g. median, IQR, histogram) of the stored parameter values for each model.

Common gaps: forgetting “with replacement” (without it, every resample is identical to the original data and there’s nothing to learn), or resampling the 5 tank-size groups instead of the 15 raw observations.

1c. No fixed correct prediction – this is meant to be tested, not answered from theory. A reasonable prediction is that the simpler, 2-parameter Ricker will be more stable (fewer parameters, less room to swing), and that’s *partly* right – but the actual result (below) is more nuanced than “simple model = stable, complex model = unstable.”

Part 2: Run the bootstrap

```
fit_ricker <- function(x, y) {
  ssr <- function(par) sum((y - par[1]*x*exp(-par[2]*x))^2)
  optim(c(1, 0.15), ssr)$par
}

fit_flex_ricker <- function(x, y) {
  ssr <- function(par) {
    a <- par[1]; p <- par[2]; c <- par[3]
    if (p <= 0 || c <= 0 || a <= 0) return(1e10)
    pred <- a * x^c * exp(-c * x / p)
    if (any(!is.finite(pred))) return(1e10)
    sum((y - pred)^2)
  }
}
```

```

}
starts <- list(c(1e-6, 12, 8), c(0.01, 12, 3), c(1, 15, 2))
best <- NULL
for (s in starts) {
  fit <- optim(s, ssr, control = list(maxit = 2000))
  if (is.null(best) || fit$value < best$value) best <- fit
}
best$par
}

set.seed(123)
nboot <- 300
n <- nrow(frogs)
ricker_boot <- matrix(NA, nboot, 2)
flex_boot <- matrix(NA, nboot, 3)

for (i in seq_len(nboot)) {
  idx <- sample(seq_len(n), size = n, replace = TRUE)
  bx <- frogs$TBL[idx]; by <- frogs$Kill[idx]
  ricker_boot[i, ] <- fit_ricker(bx, by)
  flex_boot[i, ] <- fit_flex_ricker(bx, by)
}

ricker_peak <- 1 / ricker_boot[, 2]
flex_peak <- flex_boot[, 2]
flex_shape <- flex_boot[, 3]

```

2b. Summary table (exact numbers will vary slightly by random seed and replicate count – this is one run of 300 replicates, seed 123; students’ numbers should be in the same ballpark):

```

summarize <- function(v) c(median = median(v), quantile(v, c(0.25, 0.75)))
rbind(
  `Ricker peak (1/b)`      = summarize(ricker_peak),
  `Flexible Ricker peak (p)` = summarize(flex_peak),
  `Flexible Ricker shape (c)` = summarize(flex_shape)
)

```

```

##                median      25%      75%
## Ricker peak (1/b)    6.412421  5.596696  7.240627
## Flexible Ricker peak (p) 12.996793 12.583884 13.346304
## Flexible Ricker shape (c) 20.616418 18.429946 22.207978

```

Parameter	Median	IQR
Ricker peak ($1/b$)	≈ 6.4	roughly 5.6 to 7.3
Flexible Ricker peak (p)	≈ 13.0	roughly 12.6 to 13.4
Flexible Ricker shape (c)	≈ 20 -21	roughly 18.5 to 22

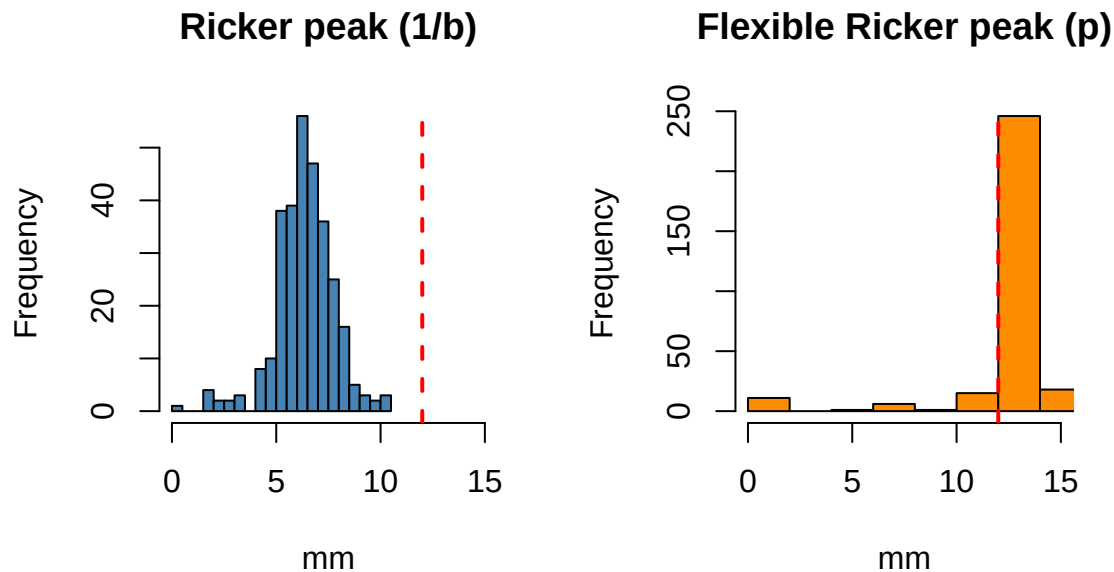
Worth noting for grading: `fit_flex_ricker()`’s a values span an enormous range across replicates (from around 10^{-22} to double digits) – if a student reports median/IQR for a too, expect it to look bizarre. That’s expected and part of the lesson, not an error.

2c. Histograms:

```

par(mfrow = c(1, 2))
hist(ricker_peak, breaks = 20, main = "Ricker peak (1/b)", xlab = "mm",
     xlim = c(0, 15), col = "steelblue")
abline(v = 12, col = "red", lwd = 2, lty = 2)
hist(flex_peak, breaks = 20, main = "Flexible Ricker peak (p)", xlab = "mm",
     xlim = c(0, 15), col = "darkorange")
abline(v = 12, col = "red", lwd = 2, lty = 2)

```



Did the plain Ricker’s bootstrapped peak locations cluster near 12 mm? No – the Ricker histogram is fairly narrow (a stable estimate) but centered around 6-7 mm, nowhere near the data’s actual peak at 12 mm. It’s precise and consistently wrong. The flexible Ricker’s peak location, by contrast, clusters tightly right around 12-13 mm – correctly located *and* reasonably stable, at least for this one parameter.

2d. This is the payoff question, and the honest answer resists a simple “model X wins.” Two things are both true and worth drawing out explicitly:

- The plain Ricker is **precise but biased**: its bootstrap distribution is fairly tight, but it’s tight around the wrong answer. You could trust its consistency, but not its accuracy.
- The flexible Ricker’s *peak location* (p) is both roughly accurate and reasonably stable – but its *shape* parameters (a and c , which control how narrow/tall the peak is) are wildly unstable across resamples. So it’s more trustworthy for “where is the peak” than for “what does the curve look like away from the peak,” or for extrapolating to sizes outside the observed range.

So: neither model is simply “the better one” – it depends on what the model needs to be used for. This is a natural place to introduce (or preview) the bias-variance tradeoff by name if it hasn’t come up yet in lecture.

Reflection. No fixed answer – use this to catch muddiest points before Wednesday’s lecture.